Responsible and Sustainable AI Report

Prepared by: Kyi Htoo Aung | Smart City Traffic Intelligence Capstone, Part 3, Task 7 | Bias and fairness, governance and sustainability of the I-94 traffic models

Summary. The traffic models are accurate on average (R^2 0.96 on unseen 2017-2018 data) but that average hides uneven performance: errors are about 57% higher in snow and storms and 2.4 times higher on holidays than on ordinary weekdays. The classification target is a **proxy** built from congestion and weather, not accident records, so it must not be used to make safety decisions. Before any real-world use the system needs validation against real incident data, a named human owner, segment-level monitoring and a documented response to alerts. The modelling footprint is deliberately small: the whole pipeline trains in about two minutes on one CPU core.

1. Bias and fairness

1.1 Sampling and coverage limitations

The models can only learn from what the single sensor recorded. The table lists the known gaps and who or what they under-represent.

Limitation	Evidence in this project	Consequence
One location, one direction	Westbound I-94 only, at one counting station between Minneapolis and St Paul	Findings and recommendations do not transfer to eastbound traffic, other roads or other cities
Gaps in time	No data from 8 Aug 2014 to 11 Jun 2015; 2012 starts in October; 2018 ends in September	Winter 2014-15 is missing entirely; seasons are unevenly represented, so yearly comparisons are biased
Broken sequences	11,842 of 40,543 hours (29%) could not be used by the LSTM because the previous 24 hours were not all recorded	The LSTM learns mainly from well-recorded periods; outage-prone periods are under-weighted
Sensor faults	32 hours on 8 days with near-zero counts at busy times were removed (Part 2)	Real closures or incidents may have been removed together with faults; the model never sees them
Rare conditions	Squall 1 hour, Smoke 2 hours, Fog 617 hours; only 431 holiday hours in the test period	Predictions for rare weather and holidays rest on very little evidence
Change of recording scheme	"Clouds" was 53% of hours in 2012 but 23% in 2015; cloud-cover values 0 and 64 largely disappear after 2014	Weather features do not mean exactly the same thing across years, which can teach the model patterns that are artefacts of labelling
Coarse weather data	One label per hour from a city-level source; snow depth is almost always recorded as 0	Road-level conditions (ice, local fog, spray) are invisible, weakening any weather-related conclusion
No data on people	Vehicle counts only: no vehicle type, trip purpose, public transport or cycling	Recommendations serve drivers; other road users and people without flexible schedules are not represented

1.2 The proxy label and its risks

No accident dataset was available, so the classifier predicts a proxy defined in the brief: **high_risk = High or Severe congestion (traffic volume above the median) AND severe weather (Thunderstorm, Squall, Snow) or low visibility (Mist, Fog, Haze, Smoke)**. It marks 7.7% of hours (7.0% of training hours, 8.9% of test hours). The risks are:

- **It is not accident data.** The label encodes an assumption (busy road plus bad weather equals danger) that was never checked against crashes. Real crash risk also depends on speed, darkness, alcohol, road works and driver behaviour, none of which are measured here.
- **It can never flag quiet periods.** By construction an hour with below-median traffic is always "low risk". Night-time, when serious crashes per vehicle are often more likely, is therefore systematically labelled safe.
- **Circularity inflates the scores.** Weather is both part of the label and a model input, so the model can rule out all clear-weather hours from its inputs alone. ROC AUC of 0.997 mostly shows that a rule can be re-learned. Scored only within severe or low-visibility hours, where the model must really predict congestion, it still reaches ROC AUC 0.985, but this measures congestion prediction, not safety.
- **Thresholds are relative.** The quartiles describe this road's own history, not capacity or danger; a different period would move the thresholds and relabel the same hour.
- **Risk of misuse.** A number called "risk" invites over-trust. If used to place patrols, set insurance prices or issue public warnings, it could misdirect resources and create false reassurance at night and in clear weather.

Mitigations applied: the label is named a proxy everywhere it appears (code, README, model registry tags, and a note returned with every API risk response); the harder within-bad-weather score is reported next to the headline score; and the recommender uses the proxy only as a small penalty, never as a safety claim.

1.3 How errors are distributed unevenly

The champion models were scored separately for each condition on the 2017-2018 test data (15,246 hours). "Median % error" is the typical error relative to the actual volume; "false-alarm rate" is the share of non-risk hours wrongly flagged.

Weather	Hours	Traffic MAE	Median % error	Risk positives	Risk recall	False-alarm rate
Clear/cloudy	10,367	225	5.2%	0	n/a	0.0%
Rain/drizzle	1,841	197	4.9%	0	n/a	0.0%
Low visibility	1,407	199	5.7%	625	97%	1.0%
Severe (snow/storm)	1,631	353	8.3%	733	98%	11.2%

Day type	Hours	Traffic MAE	Median % error	Risk positives	Risk recall	False-alarm rate
Weekday	10,440	204	4.4%	1,116	99%	0.4%
Weekend	4,375	275	8.5%	197	91%	1.1%
Holiday	431	493	13.5%	45	98%	5.4%

Time of day	Hours	Traffic MAE	Median % error	Risk positives	Risk recall	False-alarm rate
Night 0-5	3,799	90	8.2%	0	n/a	0.0%
AM peak 6-9	2,533	317	4.9%	529	99%	0.9%
Midday 10-15	3,821	246	3.5%	550	99%	1.2%
PM peak 16-19	2,546	298	4.5%	267	94%	1.5%
Evening 20-23	2,547	276	8.3%	12	33%	0.6%

Year	Hours	Traffic MAE	Median % error	Risk positives	Risk recall	False-alarm rate
2017	8,713	236	5.5%	764	98%	0.4%
2018	6,533	229	5.4%	594	97%	1.3%

- **Severe weather is predicted worst.** MAE is 353 vehicles per hour in snow and storms against 225 in clear or cloudy weather (+57%), and 11% of non-risk hours are falsely flagged. The model is least reliable exactly when travellers most need guidance. The April 2018 snowstorm (traffic 54% below forecast over two days) is the clearest example and was caught by monitoring.
- **Holidays have 2.4 times the weekday error** (MAE 493 vs 204; median error 13.5% vs 4.4%), because few holiday hours exist and each holiday behaves differently.
- **Weekends are served less well than weekdays**: higher traffic error (MAE 275) and lower risk recall (91% vs 99%). The model is tuned to the dominant weekday commuting pattern.
- **Evening risk is mostly missed**: recall is 33%, although on only 12 positive hours, so the estimate is uncertain. Night hours have a small absolute error (MAE 90) but a large relative one (8.2%).
- **Performance is stable between 2017 and 2018** (MAE 236 and 229), so there is no sign of bias growing over time, although the proxy label became more frequent (7.0% to 8.9%).

What should be done about it: report accuracy by segment, not just overall; show wider uncertainty or a caution note for severe weather and holidays; gather more examples of rare conditions or use dedicated holiday features; and keep the segment table in routine monitoring so that a decline for one group is not hidden by a good average.

2. Governance: oversight before real-world use

These models are suitable for analysis and planning support. Before they are trusted for decisions that affect the public, the following oversight should be in place.

Control	What it means here	Status in this project
Validation against reality	Test the risk output against real incident and crash records before any safety use; test forecasts on other years and locations	Not possible (no accident data); stated clearly as a limitation
Human accountability	A named owner approves each model version before it is served and can switch it off; operators decide, the model advises	Champion promotion is automatic by rule; a human approval step is needed
Traceability	Every model version, its data, parameters and metrics can be reconstructed and audited	Done: MLflow tracking and registry, MODEL_VERSIONS.md, logged pipeline, fixed random seeds
Monitoring and response	Continuous checks with a defined response to each alert	Done as a simulation: monthly error and PSI drift checks, PASS / ALERT, watch vs investigate actions

Control	What it means here	Status in this project
Transparency	Users see what the output means and its limits; explanations available	Done: proxy note in every risk response, SHAP and LIME explanations, plain-language recommendations
Security and privacy	Authentication, rate limits and input validation on the API; data minimisation	Input validation only; the mock API has no authentication. Data are aggregate counts with no personal information
Fallback and review	A safe default if the model or data feed fails; scheduled re-evaluation	Historical averages are available as a fallback; a review schedule (e.g. quarterly) should be set

A sensible release path is staged: offline use by analysts, then a shadow period in which predictions are logged but not acted on, then advisory use with human sign-off. Automated action (for example changing signs or signals) should not be based on these models without independent safety validation.

3. Sustainability: environmental and resource trade-offs

Model	Accuracy on test data	Training time	Size on disk	Comment
Ridge regression (baseline)	MAE 549, R^2 0.872	under 1 s	2 KB	Cheapest, but error is more than double
Random forest	MAE 228, R^2 0.964	4.7 s	4.4 MB	Best score by 0.001 R^2, 5 times larger
Gradient boosting (champion)	MAE 233, R^2 0.963	2.5 s	0.9 MB	Chosen: practically the same accuracy, far smaller
LSTM (next hour)	MAE 180, R^2 0.981	42 s	0.2 MB	13,249 parameters, 80 epochs
Tree model with lags (next hour)	MAE 154, R^2 0.986	about 3 s	0.4 MB	More accurate than the LSTM at a fraction of the compute

- **Small by design.** The complete Part 3 pipeline (six supervised models, clustering, rule mining, the LSTM, explanations and monitoring) runs in about 2.1 minutes on a single CPU core with no GPU. At a typical 15 to 65 W for a laptop processor that is roughly 1 to 3 watt-hours, comparable to running an LED bulb for a few minutes. This estimate is approximate and excludes the embodied footprint of the hardware.
- **Accuracy versus footprint.** Champion selection treats models within 0.005 of the best score as tied and picks the smallest, so the 0.9 MB gradient-boosting model was chosen over the 4.4 MB forest. Simpler models are also easier to explain and audit.
- **Deep learning did not earn its cost here.** The LSTM needed about 15 times the training time of the comparable tree model and was less accurate. A larger network or GPU training would add energy use with no evidence of benefit on 40,000 rows of tabular data.
- **Retraining policy matters more than one training run.** Retraining on a fixed daily schedule would multiply the footprint for no gain; retraining only when monitoring shows persistent drift is cheaper and more defensible.
- **Storage.** MLflow keeps every model version (about 22 MB here). At scale, old artifacts should be archived or pruned under a retention policy.
- **The wider balance.** The potential environmental benefit is indirect: shifting some trips out of the 07:00 and 16:00 peaks reduces stop-start driving and idling emissions. But smoother roads can also attract extra trips (rebound effect), so the system should support, not replace, public transport and demand-management policy.

4. Conclusion

The models are a credible analytical tool for understanding and forecasting demand on this corridor, built with a small computational footprint and full traceability. They are not a safety system: the risk label is a proxy, performance is weakest in severe weather and on holidays, and the data come from one sensor with known gaps. Responsible use means keeping a human in charge, reporting performance by segment, validating against real incident data before any safety-related use, and acting on monitoring alerts.